Scattering Amplitudes from Positive Geometries

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Novel Ideas in HEP: from Theory to Detector
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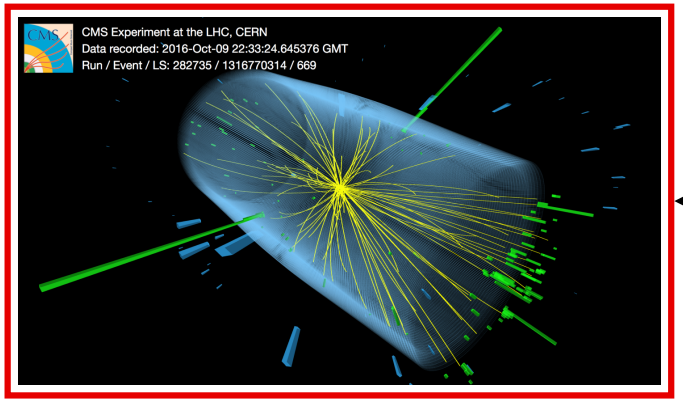
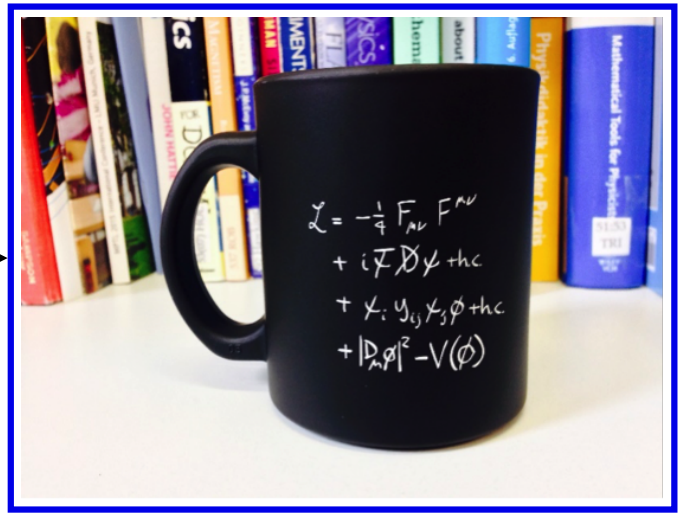
What are scattering amplitudes?

Amplitudes bridge experiment and theory.



amplitudes



calculated from theory

physical observable $\propto$ $|\text{amplitude}|^2$

- cross-sections
- decay-rates

Scattering Amplitudes and Kinematic Space

Amplitudes are functions of external data:

- *parameters*
  - masses
  - number of particles
- *quantum numbers*
  - spin ($m \neq 0$) or helicity ($m = 0$)
  - electric and/or colour charge, etc.
- *kinematic variables* (four-momenta) which are more naturally expressed in different ways for different quantum field theories (QFTs)
  - Mandelstam variables
  - spinor-helicity variables
  - momentum-twistor variables, etc.

Amplitudes have an interesting complex analytic structure in the kinematic variables.

Perturbation Theory and Feynman Diagrams

Amplitudes can be computed perturbatively using *Feynman diagrams* (FDs).

QFT specified by a Lagrangian $\mathcal{L}$

$$\downarrow$$

Feynman rules

$$\downarrow$$

$$\mathbb{A}_n = \sum_{\ell=0}^{\infty} \mathbb{A}_n^{(\ell)}$$

$$\mathbb{A}_n^{(\ell)} = \sum \underbrace{(\text{FDs with } \ell \text{ loops})}_{\text{mnemonic for an integral over } \ell \text{ variables}}$$

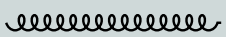
Perturbation Theory and Feynman Diagrams

Example: $SU(N_c)$ Yang-Mills (YM) Theory

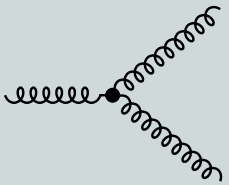
- describes the pure gluonic sector of the standard model
- gluons are massless, carry helicity and colour
- $\mathcal{L}_{\mathrm{YM}} =$

$$\mathrm{Tr}\Big[-\frac{1}{2}\boxed{(\partial_\mu A_\nu)(\partial^\nu A^\mu)} - i\sqrt{2}g\boxed{(\partial_\mu A_\nu)A^\nu A^\mu} + \frac{g^2}{4}\boxed{A_\mu A_\nu A^\mu A^\nu}\Big]$$

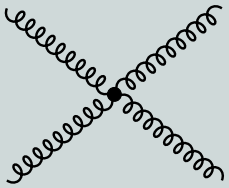
propagator



3-pt vertex

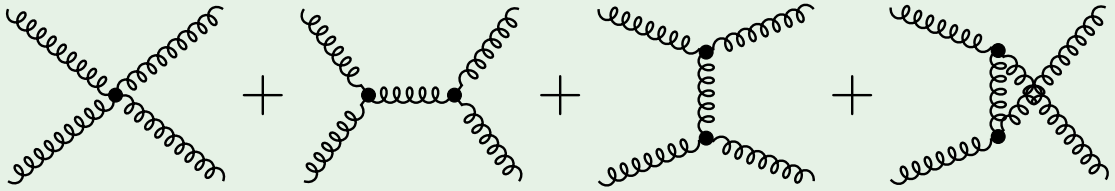


4-pt vertex



- example to $2g \to 2g$ scattering:

$\mathbb{A}_4^{(\ell=0)} =$



Perturbation Theory and Feynman Diagrams

Feynman diagrams are very successful, but they have drawbacks!

- they are not gauge-invariant
- number of diagrams grows rapidly with n:
  e.g. $2g \to (n-2)g$ at tree-level ($\ell = 0$)

n	number of diagrams
4	4
5	25
6	220
$\vdots$	$\vdots$
10	$\gtrsim$ 1 million

- complexity of individual diagrams grows with n and ℓ
- diagrams introduce redundant, *off-shell* data which hides structure and simplicity

Colour-ordered (Partial) Amplitudes

Tree-level amplitudes in non-Abelian gauge theories can be *colour-decomposed*.

Example: $SU(N_c)$ Yang-Mills (YM) Theory

$$\mathbb{A}_n^{(\ell=0)}(\{p_i, h_i, a_i\}) = g^{n-2} \sum_{\sigma \in S_n/\mathbb{Z}_n} \mathrm{Tr}\left(T^{a_{\sigma(1)}} \cdots T^{a_{\sigma(n)}}\right) \times A_n^{(\ell=0)}[\sigma(1^{h_1}, \ldots, n^{h_n})]$$

- $A_n[\sigma]$ are called *colour-ordered* amplitudes
- they depend only on kinematic variables and helicities
- they satisfy *Kleiss-Kuijf* (KK) *relations* [Kleiss and Kuijf 1989] [Del Duca, Dixon, and Maltoni 2000]

$$A_n^{(\ell=0)}[1, \{\alpha\}, n, \{\beta\}] = (-1)^{|\{\beta\}|} \sum_{\rho \in \{\alpha\} \sqcup \{\beta^T\}} A_n^{(\ell=0)}[1, \{\rho\}, n]$$

The Parke-Taylor Formula

1985 Parke and Taylor compute $2g \to 4g$ [Parke and Taylor 1986b]

- 220 FDs $\sim$ 100 pages of calculation
- result presented in 25 pages

1986 Parke and Taylor present maximal helicity violating (MHV) formula for $2g \to (n-2)g$ [Parke and Taylor 1986a]

$$A_n[1^-, 2^-, 3^+, \ldots, n^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle}$$

Spinor-Helicity Variables

appropriate kinematic variables for massless particles carrying helicity

$$p_i^{\dot{a}a} = p_i^\mu \bar{\sigma}_\mu^{\dot{a}a} = \tilde{\lambda}_i^{\dot{a}} \lambda_i^a, \qquad \langle ij \rangle = \varepsilon_{ab} \lambda_i^a \lambda_j^b, \qquad [ij] = -\varepsilon_{\dot{a}\dot{b}} \tilde{\lambda}_i^{\dot{a}} \tilde{\lambda}_j^{\dot{b}}$$

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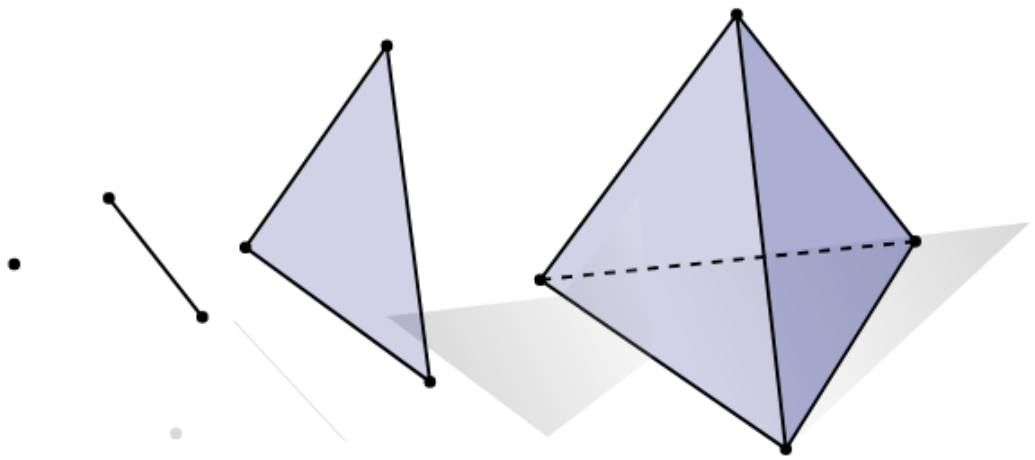
Is there a formalism which makes this structure and simplicity manifest?

Scattering Amplitudes from Positive Geometries

Positive Geometry [Arkani-Hamed, Bai, and Lam 2017]

A *positive geometry* X is a d-dimensional closed, oriented topological space with a unique rational top-form $\Omega(X)$, called the *canonical form*, satisfying the following recursive condition:

- $d > 0$: every boundary $C \subset X$ is a positive geometry of dimension $d-1$ with $\Omega(C) = \mathrm{Res}_C \Omega(X)$ and $\Omega(X)$ has no singularities elsewhere;
- $d = 0$: X is a single real point and $\Omega(X) = \pm 1$ depending on the orientation of X.



Scattering Amplitudes from Positive Geometries

Positive Geometry [Arkani-Hamed, Bai, and Lam 2017]

- 'curvy' generalizations of polytopes
- boundaries of all co-dimensions
- boundaries have a recursive structure which mimicks the factorization properties of amplitudes

For certain QFTs:

- amplitudes can be described by a positive geometry X inside the kinematic space cut out by *positivity conditions*.
- amplitudes are extracted from $\Omega(X)$
- the structure and simplicity of amplitudes are manifest in $\Omega(X)$

Scattering Amplitudes from Positive Geometries

Table of Comparisons	
Amplitudes	Positive Geometries
Amplitude $\mathbb{A}_n$	Canonical Form $\Omega(X_n)$
Physical Singularities	Boundary Structure of X_n
Factorization Channels	Residues of $\Omega(X_n)$
KK relations	?
Solutions to recursion relations	Triangulations of X_n

Scattering Amplitudes from Positive Geometries

Example: $\mathcal{N} = 4$ $SU(N_c)$ (planar) super Yang-Mills theory (pSYM)

- supersymmetric cousin of YM
- tree-level and (integrands of) loop-level amplitudes in momentum-twistor space encoded in Amplituhedron [Arkani-Hamed and Trnka 2014]
- tree-level amplitudes in spinor-helicity space encoded in Momentum Amplituhedron [Damgaard, Ferro, Łukowski, and Parisi 2019]

Scattering Amplitudes from Positive Geometries

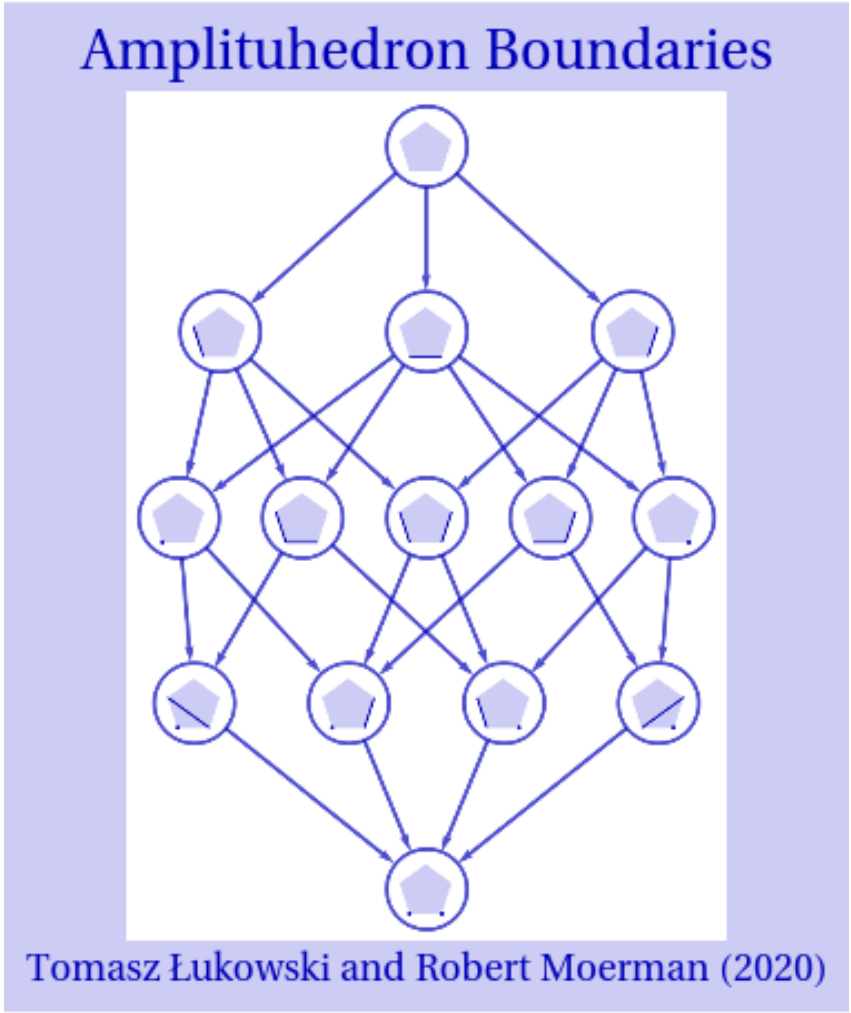
Example: $SU(N_c) \times SU(N_c)$ scalar ϕ^3 theory

- cubic interactions between massless, spinless particles each with two colours
- tree-level amplitudes encoded in Kinematic Associahedron [Arkani-Hamed, Bai, He, and Yan 2018]
- kinematic space = space of planar Mandelstam variables

PhD Research I

1. Mathematica^{TM} package `amplituhedronBoundaries.m` [Łukowski and Moerman 2021]

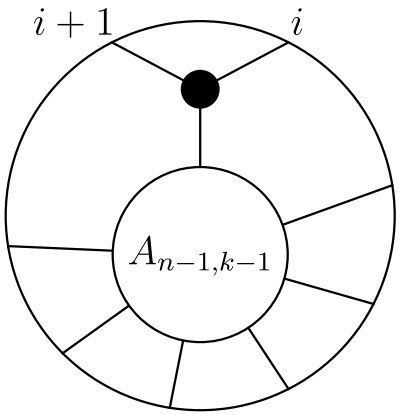
- Generates the boundary stratification of three positive geometries:
  - Amplituhedron
  - Momentum Amplituhedron
  - Hypersimplex
- Builds on the Mathematica^{TM} package `positroids.m` [Bourjaily 2012].



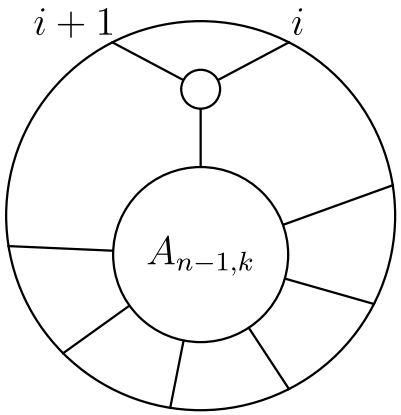
PhD Research II

2. Classification of all boundaries of the Momentum Amplituhedron [Ferro, Łukowski, and Moerman 2020]

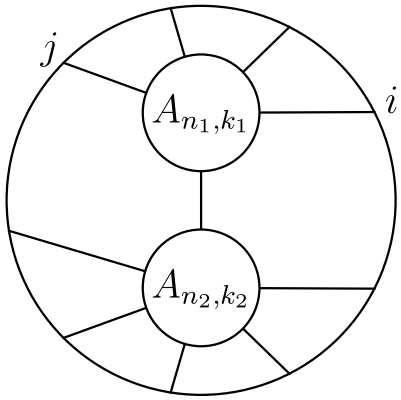
- There is a one-to-one correspondence between physical singularities of tree-level amplitudes in $\mathcal{N} = 4$ SYM and boundaries of the Momentum Amplituhedron.
- Co-dimension one boundaries encode collinear limits and factorization channels:



$[ii+1] = 0$



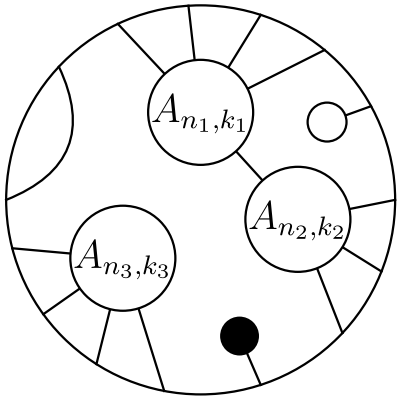
$\langle ii+1 \rangle = 0$



$s_{i,i+1,\ldots,j-1,j} = 0$

PhD Research III

- Intersecting co-dimension one boundaries produces soft limits, forward scattering limits, etc.:

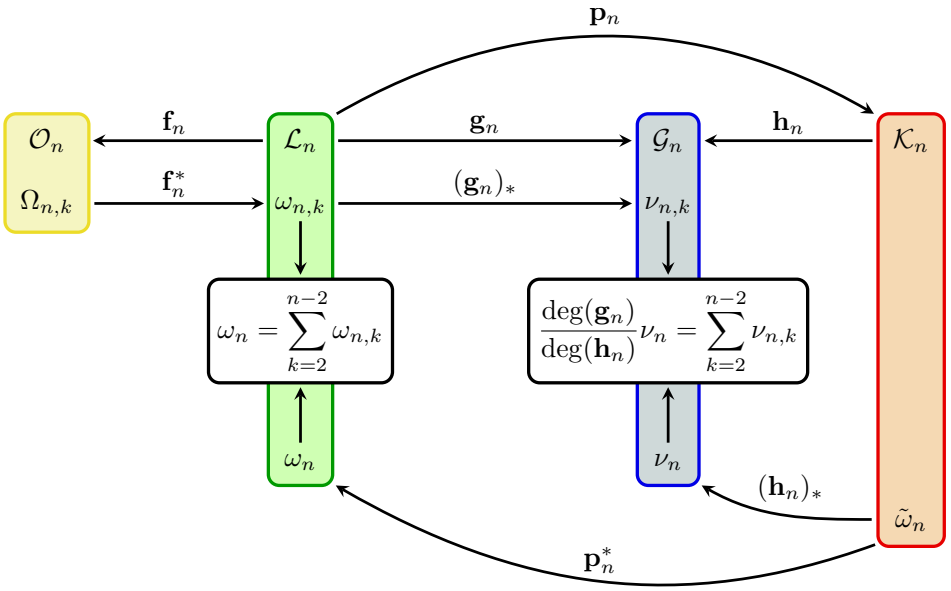


- The Momentum Amplituhedron is *Eulerian* and has *Euler Characteristic* 1, hinting that it is a closed ball.

PhD Research IV

3. A surprising connection between the Momentum Amplituhedron and the Kinematic Associahedron [Damgaard, Ferro, Łukowski, and Moerman 2021b]

- Their canonical forms $\Omega_{n,k}$ and $\tilde{\omega}_n$, respectively, live on different spaces, but they can be *pulled back/pushed forward* and compared directly.

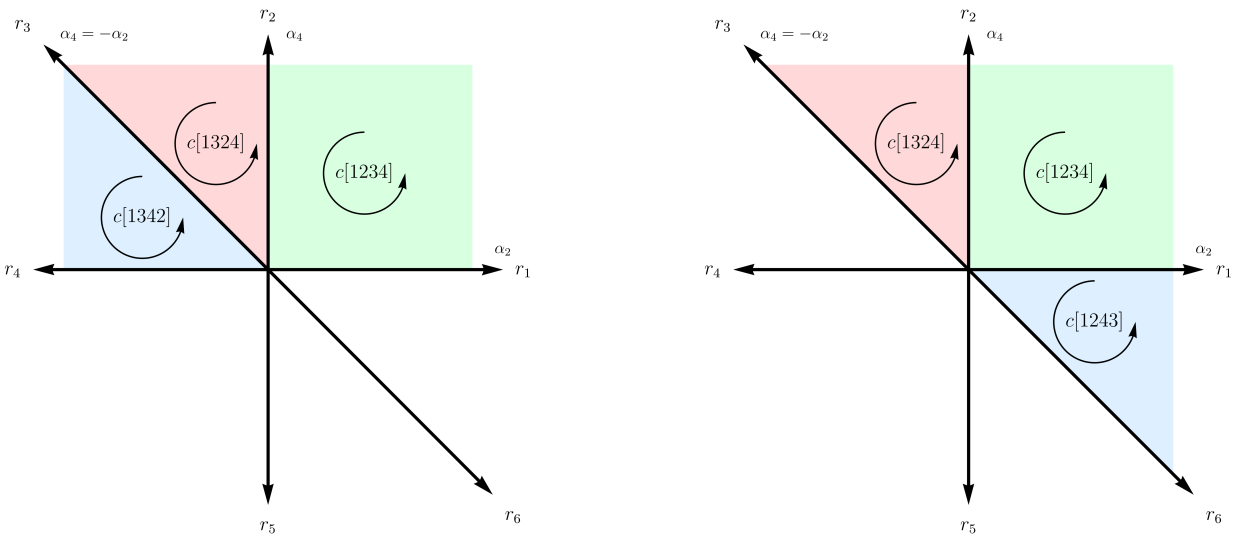


- Highlights the common singularity structure of tree-level amplitudes in both QFTs.

PhD Research V

4. Kleiss-Kuijf (KK) relations from the geometries of the Momentum Amplituhedron and the Kinematic Associahedron [Damgaard, Ferro, Łukowski, and Moerman 2021a]

- KK relations correspond to collections of positive geometries for different orderings which combine to produce a geometry without any vertices.



- KK relations can be derived from these positive geometries using a homological algorithm.

Conclusions

- amplitudes in certain quantum field theories can be described by positive geometries in kinematic space
- quantum mechanical, space-time evolution arises from 'positivity': unitarity and locality are emergent
- positive geometries make manifest the structure and simplicity of amplitudes
- boundaries of positive geometries encode the singularity structure of amplitudes
- positive geometries reveal surprising relationships between different QFTs
- 'positivity' also encodes group-theoretic information (KK relations)

Outlook

- Can we find a positive geometry for non-planar, loop-level amplitudes (or integrands thereof) in spinor-helicity space for $\mathcal{N} = 4$ SYM — a non-planar, loop-level Momentum Amplituhedron?
- Can we describe (super-)gravity amplitudes using positive geometries?
- Can we find positive geometries for more realistic theories like the standard model of particle physics?

Thank you for listening.

If you have any questions, I'd love to hear them.

References I

- Arkani-Hamed, Nima, Yuntao Bai, Song He, and Gongwang Yan (2018). "Scattering Forms and the Positive Geometry of Kinematics, Color and the Worldsheet". In: *JHEP* 05, p. 096. DOI: 10.1007/JHEP05(2018)096. arXiv: 1711.09102 [hep-th].
- Arkani-Hamed, Nima, Yuntao Bai, and Thomas Lam (2017). "Positive Geometries and Canonical Forms". In: *JHEP* 11, p. 039. DOI: 10.1007/JHEP11(2017)039. arXiv: 1703.04541 [hep-th].
- Arkani-Hamed, Nima and Jaroslav Trnka (2014). "The Amplituhedron". In: *JHEP* 10, p. 030. DOI: 10.1007/JHEP10(2014)030. arXiv: 1312.2007 [hep-th].
- Bourjaily, Jacob L. (Dec. 2012). "Positroids, Plabic Graphs, and Scattering Amplitudes in Mathematica". In: arXiv: 1212.6974 [hep-th].

References II

- Damgaard, David, Livia Ferro, Tomasz Łukowski, and Robert Moerman (Mar. 2021a). "Kleiss-Kuijf Relations from Momentum Amplituhedron Geometry". In: arXiv: 2103.13908 [hep-th].
- – (2021b). "Momentum amplituhedron meets kinematic associahedron". In: *JHEP* 02, p. 041. DOI: 10.1007/JHEP02(2021)041. arXiv: 2010.15858 [hep-th].
- Damgaard, David, Livia Ferro, Tomasz Łukowski, and Matteo Parisi (2019). "The Momentum Amplituhedron". In: *JHEP* 08, p. 042. DOI: 10.1007/JHEP08(2019)042. arXiv: 1905.04216 [hep-th].
- Del Duca, Vittorio, Lance J. Dixon, and Fabio Maltoni (2000). "New color decompositions for gauge amplitudes at tree and loop level". In: *Nucl. Phys. B* 571, pp. 51–70. DOI: 10.1016/S0550-3213(99)00809-3. arXiv: hep-ph/9910563.

References III

- Ferro, Livia, Tomasz Łukowski, and Robert Moerman (2020). "From momentum amplituhedron boundaries toamplitude singularities and back". In: *JHEP* 07.07, p. 201. DOI: 10.1007/JHEP07(2020)201. arXiv: 2003.13704 [hep-th].
- Kleiss, Ronald and Hans Kuijf (1989). "Multi - Gluon Cross-sections and Five Jet Production at Hadron Colliders". In: *Nucl. Phys. B* 312, pp. 616–644. DOI: 10.1016/0550-3213(89)90574-9.
- Łukowski, Tomasz and Robert Moerman (2021). "Boundaries of the amplituhedron with amplituhedronBoundaries". In: *Comput. Phys. Commun.* 259, p. 107653. DOI: 10.1016/j.cpc.2020.107653. arXiv: 2002.07146 [hep-th].
- Parke, Stephen J. and T. R. Taylor (1986a). "An Amplitude for n Gluon Scattering". In: *Phys. Rev. Lett.* 56, p. 2459. DOI: 10.1103/PhysRevLett.56.2459.
- – (1986b). "Gluonic Two Goes to Four". In: *Nucl. Phys. B* 269, pp. 410–420. DOI: 10.1016/0550-3213(86)90230-0.